



Properties of Logarithms

As with other mathematical functions, logarithms have their own unique properties.

Using Basic Properties of Logarithms

Aside from the basic definition of logarithms, there are three properties of logarithms that are frequently used.

Properties of Logarithms

Let $m > 0$, $n > 0$, and p be any real number.

Product Property of Logarithms

$$\log mn = \log m + \log n$$

Quotient Property of Logarithms

$$\log\left(\frac{m}{n}\right) = \log m - \log n$$

Power Property of Logarithms

$$\log(m^p) = p \log m$$

► **Think About It** All of these properties apply to logarithms with any base, including the natural logarithm, $\ln$.

EXAMPLE 1

Write as a single logarithm.

A $\log 8 + \log 4$

SOLUTION

$$\begin{aligned}\log 8 + \log 4 &= \log(8 \cdot 4) && \text{Product Property of Logarithms} \\ &= \log 32 && \text{Simplify.}\end{aligned}$$

B $\ln 15 - \ln 5 + 3 \ln 2$

SOLUTION

$$\begin{aligned}\ln 15 - \ln 5 + 3 \ln 2 &= \ln\left(\frac{15}{5}\right) + 3 \ln 2 && \text{Quotient Property of Logarithms} \\ &= \ln 3 + \ln 2^3 && \text{Power Property of Logarithms} \\ &= \ln(3 \cdot 8) && \text{Product Property of Logarithms} \\ &= \ln 24 && \text{Simplify. } \blacksquare\end{aligned}$$

EXAMPLE 2

Write in expanded form.

A $\ln\left(\frac{3}{10}\right)$

SOLUTION

Use the quotient property of logarithms.

$$\ln\left(\frac{3}{10}\right) = \ln 3 - \ln 10$$

B $\log(3x^9)$

SOLUTION

Use the product and power properties of logarithms.

$$\begin{aligned}\log(3x^9) &= \log 3 + \log(x^9) && \text{Product Property of Logarithms} \\ &= \log 3 + 9 \log x && \text{Power Property of Logarithms } \blacksquare\end{aligned}$$

Solving Logarithmic Equations

To solve equations involving logarithms, apply the properties of logarithms.

► **Think About It** If $\log a = \log b$, then $a = b$.

EXAMPLE 3

Solve for x .

A $\log x + \log 3 = \log(2x + 5)$

SOLUTION

$$\log x + \log 3 = \log(2x + 5)$$

$$\log 3x = \log(2x + 5) \quad \text{Product Property of Logarithms}$$

$$3x = 2x + 5 \quad \text{Equate arguments of logarithms.}$$

$$x = 5 \quad \text{Solve for } x.$$

B $\ln(5x - 4) = 9$

SOLUTION

$$\ln(5x - 4) = 9$$

$$e^9 = 5x - 4 \quad \text{Write in exponential form.}$$

$$e^9 + 4 = 5x$$

$$x = \frac{e^9 + 4}{5} \quad \text{Solve for } x. \quad \blacksquare$$

Application: Simplifying Formulas

EXAMPLE 4

The formula $\frac{G}{20} = \log\left(\frac{V}{K}\right)$ estimates the power gain G , in decibels, when V is the voltage being measured and K is a reference voltage. Show that $G = 20 \log V - 20 \log K$.

SOLUTION

Use the quotient property of logarithms.

$$\frac{G}{20} = \log\left(\frac{V}{K}\right)$$

$$\frac{G}{20} = \log V - \log K \quad \text{Write the right side in expanded form.}$$

$$G = 20 \log V - 20 \log K \quad \text{Multiply both sides by 20. } \blacksquare$$